Worksheet 8-9 to 8-12 A Area and Volume #1
AP Calculus Name_____

Directions: For each problem, sketch a graph of the area bounded by the given curves then find the requested information. These problems are NOT calculator active, but you may use DESMOS to graph the regions and you may use your calculator to add, subtract, multiply, or divide to get the final answer.

1.) $y = x^2,\ y = 0,\ x = 2$

a.) Find the area enclosed by the region
b.) Find the volume of the solid when the region is rotated about the $\boldsymbol{x}$**-axis**
c.) Find the volume of the solid when the region is rotated about the $\boldsymbol{y}$**-axis**
d.) Find the volume of the solid when the region is rotated about the **horizontal line** $\boldsymbol{y = 4}$
e.) Find the volume of the solid when the region is rotated about the **vertical line** $\boldsymbol{x = 2}$

2.) $y = 1 + \sqrt{x},\ y = 1,\ x = 4$

a.) Find the area enclosed by the region
b.) Find the volume of the solid when the region is rotated about the $\boldsymbol{x}$**-axis**
c.) Find the volume of the solid when the region is rotated about the **horizontal line** $\boldsymbol{y = 3}$

3.) $y = x^2$ and $y = \sqrt[3]{x}$

a.) Find the area enclosed by the region
b.) Find the volume of the solid when the region is rotated about the $\boldsymbol{x}$**-axis**
c.) Find the volume of the solid when the region is rotated about the $\boldsymbol{y}$**-axis**
d.) Find the volume of the solid when the region is rotated about the **horizontal line** $\boldsymbol{y = 1}$
e.) Find the volume of the solid when the region is rotated about the **vertical line** $\boldsymbol{x = 1}$

Answers:

1.) (a) $\frac{8}{3}$ (b) $\frac{32\pi}{5}$ (c) 8π (d) $\frac{224\pi}{15}$ (e) $\frac{8\pi}{3}$

2.) (a) $\frac{16}{3}$ (b) $\frac{56\pi}{3}$ (c) $\frac{40\pi}{3}$

3.) (a) $\frac{5}{12}$ (b) $\frac{2\pi}{5}$ (c) $\frac{5\pi}{14}$ (d) $\frac{13\pi}{30}$ (e) $\frac{10\pi}{21}$